

Decision Tree

Algorithm	criterion	max_depth	max_features	min_samples_leaf	min_samples_split	splitter	R2 Score
Decision Tree Regression	squared_error	5		2	2	best	0.88022
Decision Tree Regression	squared_error	5		2	5	best	0.88022
Decision Tree Regression	friedman_mse	5		2	5	best	0.88022
Decision Tree Regression	friedman_mse	5		2	2	best	0.88022
Decision Tree Regression	squared_error	5	sqrt	2	5	best	0.864142
Decision Tree Regression	friedman_mse	5	sqrt	2	5	best	0.864142
Decision Tree Regression	friedman_mse	5	sqrt	2	2	best	0.864142
Decision Tree Regression	friedman_mse	5	sqrt	1	5	best	0.864142
Decision Tree Regression	squared_error	5	sqrt	1	5	best	0.864142
Decision Tree Regression	squared_error	5	sqrt	2	2	best	0.864142
Decision Tree Regression	friedman_mse	10		2	5	random	0.850533
Decision Tree Regression	squared_error	10		2	5	random	0.850533
Decision Tree Regression	squared_error	5		1	2	best	0.842067
Decision Tree Regression	friedman_mse	5		1	5	best	0.842067
Decision Tree Regression	friedman_mse	5		1	2	best	0.842067
Decision Tree Regression	squared_error	5		1	5	best	0.842067
Decision Tree Regression	friedman_mse	15		2	2	random	0.837979
Decision Tree Regression	squared_error	15		2	2	random	0.837979
Decision Tree Regression	friedman_mse	10		2	2	best	0.837223
Decision Tree Regression	squared_error	10		2	2	best	0.837223
Decision Tree Regression	squared_error	10		2	5	best	0.837061
Decision Tree Regression	friedman_mse	10		2	5	best	0.83705
Decision Tree Regression	squared_error	15		2	5	best	0.830321
Decision Tree Regression	friedman_mse	15		2	5	best	0.830321
Decision Tree Regression	squared_error	15		2	2	best	0.827515
Decision Tree Regression	friedman_mse	15		2	2	best	0.8273
Decision Tree Regression	squared_error	15	sqrt	2	2	best	0.827096
Decision Tree Regression	friedman_mse	15	sqrt	2	2	best	0.827096
Decision Tree Regression	squared_error	5	sqrt	1	2	best	0.825195
Decision Tree Regression	friedman_mse	5	sqrt	1	2	best	0.825195
Decision Tree Regression	friedman_mse	15		2	5	random	0.819812

Multiple Linear Regression

Algorithm	fit_intercept	copy_X	positive	tol	R2 Score
Multiple Linear Regression	TRUE	TRUE	FALSE	1.00E-03	0.789479
Multiple Linear Regression	TRUE	TRUE	FALSE	1.00E-06	0.789479
Multiple Linear Regression	TRUE	FALSE	FALSE	1.00E-03	0.789479
Multiple Linear Regression	TRUE	FALSE	FALSE	1.00E-06	0.789479
Multiple Linear Regression	TRUE	TRUE	TRUE	1.00E-03	0.789443
Multiple Linear Regression	TRUE	TRUE	TRUE	1.00E-06	0.789443
Multiple Linear Regression	TRUE	FALSE	TRUE	1.00E-03	0.789443
Multiple Linear Regression	TRUE	FALSE	TRUE	1.00E-06	0.789443
Multiple Linear Regression	FALSE	TRUE	FALSE	1.00E-03	0.762372
Multiple Linear Regression	FALSE	TRUE	FALSE	1.00E-06	0.762372
Multiple Linear Regression	FALSE	FALSE	FALSE	1.00E-03	0.762372
Multiple Linear Regression	FALSE	FALSE	FALSE	1.00E-06	0.762372
Multiple Linear Regression	FALSE	TRUE	TRUE	1.00E-03	0.761433
Multiple Linear Regression	FALSE	TRUE	TRUE	1.00E-06	0.761433
Multiple Linear Regression	FALSE	FALSE	TRUE	1.00E-03	0.761433
Multiple Linear Regression	FALSE	FALSE	TRUE	1.00E-06	0.761433

Support Vector Machine

Algorithm	kernel	C	gamma	degree	coef0	epsilon	R2 Score
Support Vector Regression	linear	10				0.1	0.462468414
Support Vector Regression	linear	10				0.5	0.462460209
Support Vector Regression	sigmoid	10	auto		0	0.5	0.039307144
Support Vector Regression	sigmoid	10	scale		0	0.1	0.039307144
Support Vector Regression	sigmoid	10	scale		0	0.5	0.039307144
Support Vector Regression	sigmoid	10	auto		0	0.1	0.039307144
Support Vector Regression	poly	10	scale	3		0.1	0.038716223
Support Vector Regression	poly	10	auto	3		0.1	0.038716223
Support Vector Regression	poly	10	auto	3		0.5	0.038701145
Support Vector Regression	poly	10	scale	3		0.5	0.038701145
Support Vector Regression	poly	10	scale	4		0.1	0.009871581
Support Vector Regression	poly	10	auto	4		0.1	0.009871581
Support Vector Regression	poly	10	scale	4		0.5	0.009854078
Support Vector Regression	poly	10	auto	4		0.5	0.009854078
Support Vector Regression	linear	1				0.5	-0.010102665
Support Vector Regression	linear	1				0.1	-0.010102665
Support Vector Regression	rbf	10	0.1			0.5	-0.028197712
Support Vector Regression	rbf	10	0.1			0.1	-0.028197712
Support Vector Regression	sigmoid	10	auto		1	0.5	-0.029313279
Support Vector Regression	sigmoid	10	scale		1	0.1	-0.029313279
Support Vector Regression	sigmoid	10	scale		1	0.5	-0.029313279
Support Vector Regression	sigmoid	10	auto		1	0.1	-0.029313279
Support Vector Regression	rbf	10	scale			0.1	-0.032273294
Support Vector Regression	rbf	10	auto			0.1	-0.032273294
Support Vector Regression	rbf	10	scale			0.5	-0.032273634
Support Vector Regression	rbf	10	auto			0.5	-0.032273634
Support Vector Regression	poly	10	scale	2		0.1	-0.032303464
Support Vector Regression	poly	10	auto	2		0.1	-0.032303464
Support Vector Regression	poly	10	scale	2		0.5	-0.032324794
Support Vector Regression	poly	10	auto	2		0.5	-0.032324794
Support Vector Regression	rbf	10	0.01			0.5	-0.074019412

Random Forest

Algorithm	bootstrap	criterion	max_depth	max_features	min_samples_leaf	min_samples_split	n_estimators	R2 Score
Random Forest Regression	TRUE	friedman_mse	8	sqrt	2	2	300	0.886186
Random Forest Regression	TRUE	friedman_mse	8	sqrt	2	3	300	0.886186
Random Forest Regression	TRUE	squared_error	8	sqrt	2	3	300	0.88617
Random Forest Regression	TRUE	squared_error	8	sqrt	2	2	300	0.88617
Random Forest Regression	TRUE	squared_error	8	sqrt	3	2	200	0.885149
Random Forest Regression	TRUE	squared_error	8	sqrt	3	3	200	0.885149
Random Forest Regression	TRUE	friedman_mse	8	sqrt	3	3	200	0.885149
Random Forest Regression	TRUE	friedman_mse	8	sqrt	3	2	200	0.885149
Random Forest Regression	TRUE	friedman_mse	8	sqrt	3	3	300	0.884684
Random Forest Regression	TRUE	friedman_mse	8	sqrt	3	2	300	0.884684
Random Forest Regression	TRUE	squared_error	8	sqrt	3	2	300	0.884678
Random Forest Regression	TRUE	squared_error	8	sqrt	3	3	300	0.884678
Random Forest Regression	TRUE	friedman_mse	8	sqrt	2	3	200	0.884311
Random Forest Regression	TRUE	friedman_mse	8	sqrt	2	2	200	0.884311
Random Forest Regression	TRUE	squared_error	8	sqrt	2	2	200	0.88429
Random Forest Regression	TRUE	squared_error	8	sqrt	2	3	200	0.88429
Random Forest Regression	FALSE	squared_error	8	sqrt	2	3	200	0.883568
Random Forest Regression	FALSE	squared_error	8	sqrt	2	2	200	0.883568
Random Forest Regression	FALSE	friedman_mse	8	sqrt	2	3	200	0.883505
Random Forest Regression	FALSE	friedman_mse	8	sqrt	2	2	200	0.883505
Random Forest Regression	FALSE	squared_error	8	sqrt	3	2	300	0.883502
Random Forest Regression	FALSE	squared_error	8	sqrt	3	3	300	0.883502
Random Forest Regression	FALSE	friedman_mse	8	sqrt	3	2	300	0.8835
Random Forest Regression	FALSE	friedman_mse	8	sqrt	3	3	300	0.8835
Random Forest Regression	TRUE	squared_error	10	sqrt	3	3	300	0.883443